A Low-Cost SDR-Edge Architecture for Weather Telemetry in Cyber-Physical Greenhouse Drying Systems

Abstract: This study presents a low-cost edge architecture based on software-defined radio for receiving, validating, decoding and storing proprietary weather telemetry in agricultural cyber-physical systems. The system was implemented on an ARM/Linux platform with an RTL-SDR receiver and integrates cyclic redundancy checking, asynchronous processing, local backup logging, embedded database persistence and watchdog-based automatic recovery. Decoded variables were validated against a Davis reference system using 286 time-matched observations. Air temperature and relative humidity achieved mean absolute errors of 0.156 °C and 0.352%, respectively, with correlation coefficients above 0.99. Solar irradiance and ultraviolet index produced correlation coefficients of 0.951 and 0.974. During a six-day field deployment, the architecture stored 8,612 validated records, achieved a packet delivery ratio of 95.10%, a mean recovery time of 1.0 s, and stable one-minute data persistence. The results demonstrate reliable and interoperable weather telemetry acquisition for agricultural edge-computing applications

Keywords: software-defined radio; weather telemetry; cyber-physical systems; edge computing; greenhouse drying; precision agriculture; interoperability; reliability engineering

1 Introduction

The automation of protected agriculture has shifted from isolated sensing systems towards cyber-physical systems (CPS), in which embedded computation, communication networks, environmental sensing and physical actuation are integrated for real-time decision-making (Lee, 2008; Gubbi et al., 2013; Tzounis et al., 2017; An et al., 2017). Agricultural CPS are now regarded as a technological framework for sustainable, data-driven and adaptive production systems.

In fruit-drying greenhouses, a purely internal sensing strategy is often insufficient. If the internal relative humidity is high, a reactive controller may open vents or activate extractors; yet when the external air is saturated or rain is falling, ventilation reintroduces moisture and degrades product quality. Access to external meteorological data is therefore a prerequisite for proactive control in low-cost drying systems intended for small-scale producers. Greenhouse dryers are well suited to regional agro-industrial applications because they reduce exposure to dust, rainfall, insects and animals while using solar energy as the main thermal source (Prakash and Kumar, 2014; Ahmad et al., 2022). The role of weather monitoring in digital agriculture has been documented in studies of agricultural sensor networks, precision agriculture and low-cost agrometeorological stations, where real-time environmental information is a primary input for operational decisions and risk management (Wang et al., 2006; Pierce and Elliott, 2008; Majumdar et al., 2021).

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Professional weather stations provide high-quality measurements, but their integration with custom controllers is often constrained by proprietary radio protocols or closed commercial interfaces. Software-defined radio (SDR) offers an alternative by moving part of the radiofrequency processing chain into software, which allows low-cost reception, analysis and reconfiguration of telemetry signals. SDR has consequently become relevant for spectrum monitoring, environmental sensing and research instrumentation, since it gives access to radiofrequency information without application-specific hardware (Mitola, 1995; Doha and Abdelhadi, 2026; Machado and Wyglinski, 2015; Sorecau et al., 2023).

This study contributes the design and field evaluation of a low-cost SDR–edge architecture that receives, validates, decodes and stores weather telemetry in real time under outdoor conditions. Unlike a data logger, the proposed architecture adds CRC validation, bit-level decoding of multiplexed variables, asynchronous processing, redundant local and database storage of time-stamped records, and a watchdog-based auto-recovery mechanism that improves availability on resource-constrained hardware. The architecture thus addresses the requirements of IoT-enabled protected agriculture, where local sensing, edge processing, decision support and interoperability must coexist in a single deployment (Maraveas and Bartzanas, 2021; Sacaleanu et al., 2024).

The specific objectives were: (i) to implement an SDR-based receiver for weather telemetry in the 902–928 MHz industrial, scientific and medical (ISM) band; (ii) to decode meteorological variables from validated payloads; (iii) to structure the data flow for edge storage and future controller access; and (iv) to quantify reliability through the packet delivery ratio, the mean time to recovery (MTTR), the database-writing jitter and Weibull-based failure behaviour. Agricultural IoT, weather monitoring and edge-computing platforms have been studied extensively, but less attention has been paid to low-cost SDR receivers used to decode proprietary weather telemetry, to their interoperability with commercial sensing platforms, and to the quantification of operational reliability through metrics such as the mean time between failures (MTBF), MTTR and Weibull failure modelling.

2 Related Work

2.1 IoT and CPS in protected agriculture

IoT-based monitoring is now common in protected agriculture because it supports continuous acquisition of microclimatic information, remote visualisation and automated decision support (Akyildiz et al., 2002; Gubbi et al., 2013; Tzounis et al., 2017; Maraveas and Bartzanas, 2021; Ferentinos et al., 2017). Smart-farming studies have examined sensor networks, data analytics, distributed decision support and low-cost monitoring infrastructures as means of improving productivity and sustainability (Wolfert et al., 2017; Liakos et al., 2018; Wang et al., 2006; Pierce and Elliott, 2008; Majumdar et al., 2021). Agricultural CPS, however, require more than data visualisation: they require reliable data flows, synchronisation between physical and digital states, and mechanisms that tolerate faults in field-deployed hardware (Lee, 2008; Shi et al., 2016; Qian et al., 2024). These requirements are stringent in drying applications, where the process responds directly to temperature and humidity excursions (Gorjian et al., 2021; Ahmad et al., 2022). Recent agricultural CPS research has emphasised the integration of sensing, communication,

computation and decision-making as the basis for autonomous and adaptive systems (An et al., 2017).

2.2 SDR for environmental telemetry

SDR provides a reconfigurable platform for receiving, demodulating and analysing radio signals through software workflows (Mitola, 1995; Doha and Abdelhadi, 2026; Machado and Wyglinski, 2015). SDR-based approaches have also been explored for environmental sensing and distributed monitoring beyond conventional communication systems (Salam et al., 2019). Low-cost RTL-SDR devices are widely used in educational, experimental and monitoring settings because they give access to radiofrequency signals without a dedicated proprietary receiver. For weather telemetry, SDR enables interoperability when commercial sensors transmit wirelessly but expose no open digital interface. Community-developed tools show that telemetry from Davis weather stations can be recovered by SDR: the `rtldavis` receiver (Hall, 2015), later extended with multi-channel frequency hopping (Heijst, 2019), and the `weewx-rtldavis` driver that couples that receiver to the WeeWX weather-station software (Wall and Heijst, 2019). Further work is nevertheless needed to turn decoded packets into fault-tolerant data streams for agricultural CPS. SDR has likewise been applied to spectrum monitoring, environmental sensing and climate-related observation, which illustrates its versatility as a low-cost instrumentation platform (Sorecau et al., 2023; Guevara et al., 2025).

2.3 Edge computing and local data management

Edge computing suits agricultural environments in which connectivity is intermittent and local decisions must continue when cloud services are unavailable (Shi et al., 2016; Gong et al., 2025; Yu et al., 2025). Environmental monitoring requires a store that organises time-stamped observations efficiently and lets applications, dashboards and controllers query recent or historical states; on resource-constrained edge nodes an embedded SQL engine meets this requirement without a database server, whereas dedicated time-series engines become advantageous once several nodes or long retention periods are involved. Combining SDR reception with edge validation and local time-stamped storage therefore supports local decision-making in small-scale protected agriculture, in line with decision-support approaches that couple sensor data, software platforms and user-oriented interfaces to guide resource-efficient management (Yousaf et al., 2022; Petraki et al., 2025; Sacaleanu et al., 2024).

2.4 Reliability and fault tolerance in agricultural monitoring systems

Reliability and fault tolerance have gained attention in agricultural IoT and CPS deployments because monitoring infrastructures operate outdoors and with intermittent connectivity. Published work covers sensor-fault detection, anomaly diagnosis, redundancy mechanisms and resilient software architectures for agricultural environments (Shekarian et al., 2024; Alotaibi, 2026). Reliability evaluation, however, usually focuses on sensing performance or communication availability, whereas operational metrics such as MTBF, MTTR and Weibull reliability modelling are rarely reported for agricultural telemetry systems.

3 Materials and Methods

3.1 Application context

The field evaluation was conducted in Oaxaca de Juárez, Oaxaca, Mexico (17°03’38” N, 96°43’31” W), at an elevation of approximately 1,550 m above mean sea level. Weather telemetry was transmitted by a Davis Vantage Pro2 wireless weather station and received using an SDR-based edge acquisition system installed approximately 15 m from the transmitting station throughout the experimental campaign. This short separation distance was selected to evaluate decoding fidelity, packet integrity, and operational reliability under field conditions while minimising propagation-related losses. The external weather acquisition layer was designed as the first stage of a CPS in which outdoor meteorological conditions are combined with internal chamber measurements to support the future control of ventilation, heating and vent position. Figure 1 summarises the acquisition, validation, storage and control workflow.

3.2 Hardware platform and weather telemetry reception

The acquisition system was implemented using a Nooelec NESDR SMArt software-defined radio receiver (RTL2832U and R820T2 chipset, 0.5 PPM temperature-compensated crystal oscillator, SMA RF input) connected to a Raspberry Pi 5 single-board computer equipped with 16 GB RAM and running Ubuntu Linux on the `aarch64` architecture. RF reception was performed using a 915 MHz fibreglass LoRa antenna connected through a low-loss RG58 coaxial cable. The telemetry source was a Davis Vantage Pro2 wireless weather station operating in the 902–928 MHz ISM band. The SDR platform was selected to provide low-cost interception and decoding of proprietary weather telemetry while maintaining sufficient frequency stability for long-duration operation.

Radiofrequency reception and channel tracking were performed by the open-source `rtldavis` receiver (Hall, 2015; Heijst, 2019), operated through the `weewx-rtldavis` driver for the WeeWX weather-station software (Wall and Heijst, 2019), configured for the US hop-frequency set of the 902–928 MHz band. Both components are distributed under the GNU General Public License v3. The weather station transmits in the 902–928 MHz ISM

Table 1 Hardware components used during the field evaluation.

Component	Description
Weather station	Davis Vantage Pro2
SDR receiver	Nooelec NESDR SMArt
RF chipset	RTL2832U + R820T2
Frequency stability	0.5 PPM TCXO
Receiving antenna	915 MHz fibreglass LoRa antenna
Edge computer	Raspberry Pi 5 (16 GB RAM)
Operating system	Ubuntu Linux (`aarch64`)
Station-receiver distance	15 m

band using frequency-hopping spread spectrum (FHSS), hopping between channels every few seconds. The receiver software follows this radio activity and extracts the hexadecimal payload of each valid telemetry burst.

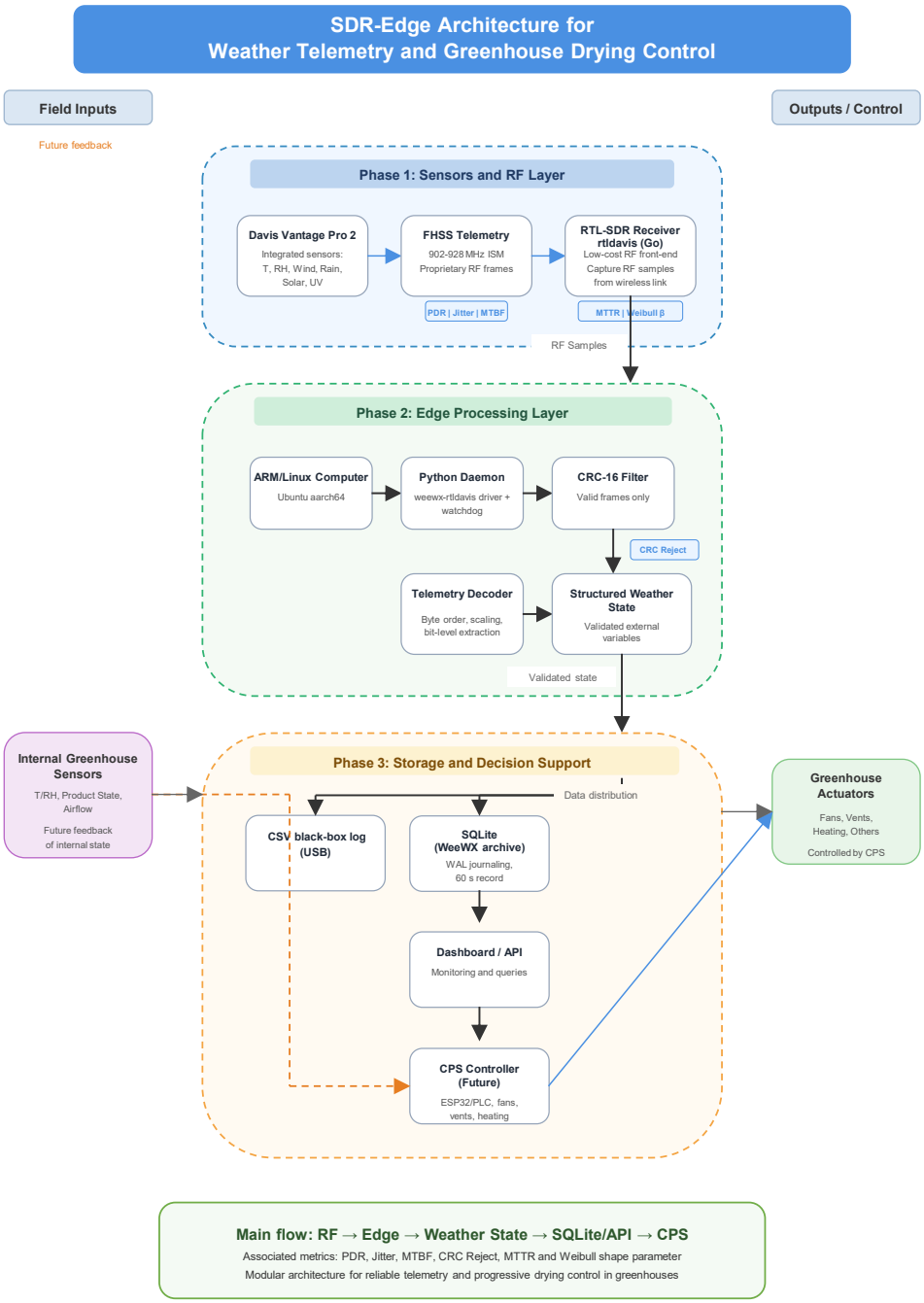


Figure 1 Proposed SDR-based edge architecture for weather-telemetry decoding and future greenhouse drying control. The architecture links the Davis integrated sensor suite and its FHSS radio telemetry with an RTL-SDR receiver, an ARM/Linux edge computer, a Python daemon under watchdog supervision, CRC-16 validation, variable decoding, redundant local CSV logging, persistence in a SQLite archive, a dashboard/API layer and future greenhouse actuators.

3.3 Packet structure and CRC validation

Each intercepted packet was treated as an 8-byte frame: the last two bytes carry the CRC-16 checksum and the first six bytes carry the payload used for variable decoding. Frames whose computed CRC did not match the received value were discarded, so that corrupted measurements never reached the database or the future control logic.

The packet delivery ratio (PDR) was computed as

$$\mathrm{PDR}(\%) = \frac{N_{\mathrm{valid}}}{N_{\mathrm{valid}} + N_{\mathrm{CRC\ errors}}} \times 100, \tag{1}$$

where N_{valid} is the number of accepted packets and $N_{\mathrm{CRC\ errors}}$ the number rejected by the CRC filter.

3.4 Decoding of meteorological variables

Wind speed and wind direction occupy the fixed portion of the telemetry frame. Wind direction was mapped from an 8-bit integer to degrees using

$$\theta = \frac{x}{255} \times 360, \tag{2}$$

where x is the decoded byte value.

Temperature and relative humidity were decoded from multiplexed payload bytes. Temperature was interpreted as a signed 16-bit value with scaling and unit conversion, whereas relative humidity required fractional byte ordering and 12-bit extraction. Solar irradiance and UV index were decoded with bitwise operations over compact 10-bit fields. The general extraction form used for the optical variables was

$$V_{10} = ((b_3 \ll 2) \mid (b_4 \gg 6)) \ \&\ \texttt{0x3FF}, \tag{3}$$

where b_3 and b_4 are payload bytes, $\ll$ and $\gg$ denote left and right bit shifts, $\mid$ is the bitwise OR and $\&$ applies a 10-bit mask.

Table 2 lists the decoded variables, the interpretation strategy applied to each and their relevance for greenhouse drying decisions. The payload structure and the field-extraction rules summarised above follow the open-source implementation of `rtldavis` and `weewx-rtldavis` (Hall, 2015; Heijst, 2019; Wall and Heijst, 2019); the contribution of the present work lies in the layer built around them, namely the explicit CRC accounting, the asynchronous daemon with watchdog supervision, the redundant persistence path and the reliability instrumentation described below.

3.5 Software architecture

The software was refactored from a linear prototype into an asynchronous daemon. One thread is dedicated to SDR listening, while the main loop handles screen updates, data formatting and persistence. Shared data structures are protected by locks to avoid race conditions, and circular buffers hold short-term trends so that memory use remains bounded.

Table 2 Decoded weather variables and relevance for greenhouse drying decision support.

Variable	Decoding approach	Relevance for greenhouse drying
Air temperature	Signed 16-bit conversion with scaling and unit transformation	External thermal potential available for solar-assisted drying and ventilation decisions.
Relative humidity	Multiplexed 12-bit extraction with fractional byte ordering	Whether outside air will remove or introduce moisture in the drying chamber.
Solar irradiance	10-bit bitwise extraction from compact optical payload fields	Solar-energy availability for greenhouse heating and drying.
UV index	10-bit bitwise extraction and scale conversion	Additional optical consistency indicator for daylight and cloud transients.
Wind speed and direction	Fixed payload fields decoded from recurrent telemetry frames	Risk assessment for ventilation, air exchange and external exposure.
Rain state/rate	Rain-related payload interpretation and dry-state marker handling	Basis for inhibiting vent opening during rainfall or high external moisture.

3.6 Data persistence

The pipeline uses two persistence mechanisms. A local CSV file acts as a redundant black-box log in case of database failure, and the validated, decoded variables are written to the WeeWX SQLite archive at nominal 60 s intervals. SQLite was operated in write-ahead logging (WAL) mode, which keeps readers from blocking the writer, so dashboards and future controllers can query the latest external weather state without interrupting SDR reception. An embedded SQL engine was preferred over a database server because it requires no additional service on the edge node and keeps the whole archive in a single file that is simple to back up in the field. Comparable data-management strategies have been adopted in agricultural monitoring infrastructures that require robust field operation and long-term data availability (Fisher et al., 2018).

3.7 Validation against the Davis reference system

The meteorological variables reconstructed from the intercepted RF packets were compared with the observations recorded by the Davis Vantage Pro2 reference system. A total of 286 time-matched observation pairs were obtained by synchronising both datasets with a maximum tolerance window of 5 min. Air temperature, relative humidity, solar irradiance and UV index were validated. Agreement was quantified through the mean absolute error (MAE), the root mean square error (RMSE), the mean bias, Pearson's correlation coefficient (r) and the coefficient of determination (R^2):

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} \left|\mathrm{SDR}_i - \mathrm{Davis}_i\right|, \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\mathrm{SDR}_i - \mathrm{Davis}_i\right)^2}, \tag{5}$$

$$\mathrm{Bias} = \frac{1}{n}\sum_{i=1}^{n}\left(\mathrm{SDR}_i - \mathrm{Davis}_i\right), \tag{6}$$

where SDR_i and Davis_i are paired observations from the SDR receiver and the Davis system. Scatter plots with identity lines and Bland–Altman diagrams were additionally used to inspect agreement across the full range of environmental conditions and to identify systematic offsets. This procedure assesses the fidelity of the SDR decoding process relative to the manufacturer's interpretation of the same measurements; it is not a metrological calibration against independently traceable instruments.

3.8 Watchdog and reliability metrics

A watchdog timer was implemented to detect receiver stalls: if no radiofrequency activity was detected during a 30 s window, the SDR subprocess was terminated and restarted. The mean time to recovery was computed as

$$\mathrm{MTTR} = \frac{1}{n}\sum_{i=1}^{n} t_{r,i}, \tag{7}$$

where $t_{r,i}$ is the recovery time of the i-th watchdog intervention.

Database-writing jitter was quantified as the standard deviation of the intervals between consecutive writes,

$$J = \sqrt{\frac{\sum_{i=1}^{n}(\Delta t_i - \overline{\Delta t})^2}{n-1}}. \tag{8}$$

For long-term reliability interpretation, the times between receiver stalls were fitted to a Weibull distribution,

$$f(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-(t/\eta)^{\beta}}, \tag{9}$$

where β is the shape parameter and η the scale parameter (Alotaibi, 2026). Both parameters were estimated by maximum likelihood from the observed times between receiver stalls.

4 Results

4.1 Validation against Davis measurements

The purpose of the validation phase was to establish whether the SDR decoding workflow reproduced the environmental information carried by the proprietary Davis telemetry link. Table 3 reports the agreement statistics for the 286 time-matched observations. Air

temperature and relative humidity agreed with the reference system to within a MAE of 0.156 °C and 0.352%, with correlation coefficients above 0.99 and biases of 0.011 °C and -0.017%. Solar irradiance and UV index gave correlation coefficients of 0.951 and 0.974. The larger error metrics of the two optical variables are consistent with temporal aggregation over the 5 min matching window, during which irradiance can change rapidly, rather than with errors in the bit-level decoding. Taken together, the statistics indicate that the architecture reconstructs the information transmitted by the Davis station with a resolution adequate for agricultural decision support.

Table 3 Agreement statistics between SDR-decoded measurements and the Davis reference system ($n = 286$ matched pairs).

Variable	MAE	RMSE	Bias	r	R^2
Temperature (°C)	0.156	0.181	0.011	0.999	0.998
Relative humidity (%)	0.352	0.495	−0.017	0.999	0.999
Solar irradiance ($\mathrm{W\,m^{-2}}$)	39.329	87.420	0.118	0.951	0.904
UV index	0.333	0.742	0.040	0.974	0.950

4.2 Agreement analysis using Bland–Altman plots

To complement the error metrics and correlation analysis, Bland–Altman plots were constructed for air temperature, relative humidity, solar irradiance, and UV index (Figure 2). This analysis evaluates measurement agreement by representing the difference between SDR-derived and Davis observations against their mean value. Temperature and relative humidity exhibited narrow limits of agreement and negligible systematic bias, indicating that the SDR decoding workflow reproduced the original telemetry values with high consistency across the observed measurement ranges. Solar irradiance and UV index exhibited broader limits of agreement. However, the differences remained centred around zero bias, supporting the interpretation that the observed deviations primarily originated from differences in temporal aggregation and measurement dynamics rather than systematic decoding errors. Overall, the Bland–Altman analysis corroborates the quantitative validation metrics and supports the conclusion that the SDR-based architecture accurately reconstructs the environmental information transmitted by the Davis weather station.

4.3 Telemetry decoding and data availability

During the field evaluation, from 29 August to 4 September 2026, the system stored 8,612 validated database records. No permanent software failure occurred, and the edge architecture sustained acquisition and persistence of the decoded variables throughout the period. Table 4 summarises the reliability and performance indicators obtained during the test.

4.4 RF link quality

More than 230,000 frames passed CRC-16 validation during the evaluation period, and 11,899 were rejected as corrupted or incomplete. The resulting packet delivery ratio was 95.10%, so most intercepted frames were usable for variable reconstruction despite operation in an unlicensed ISM band.

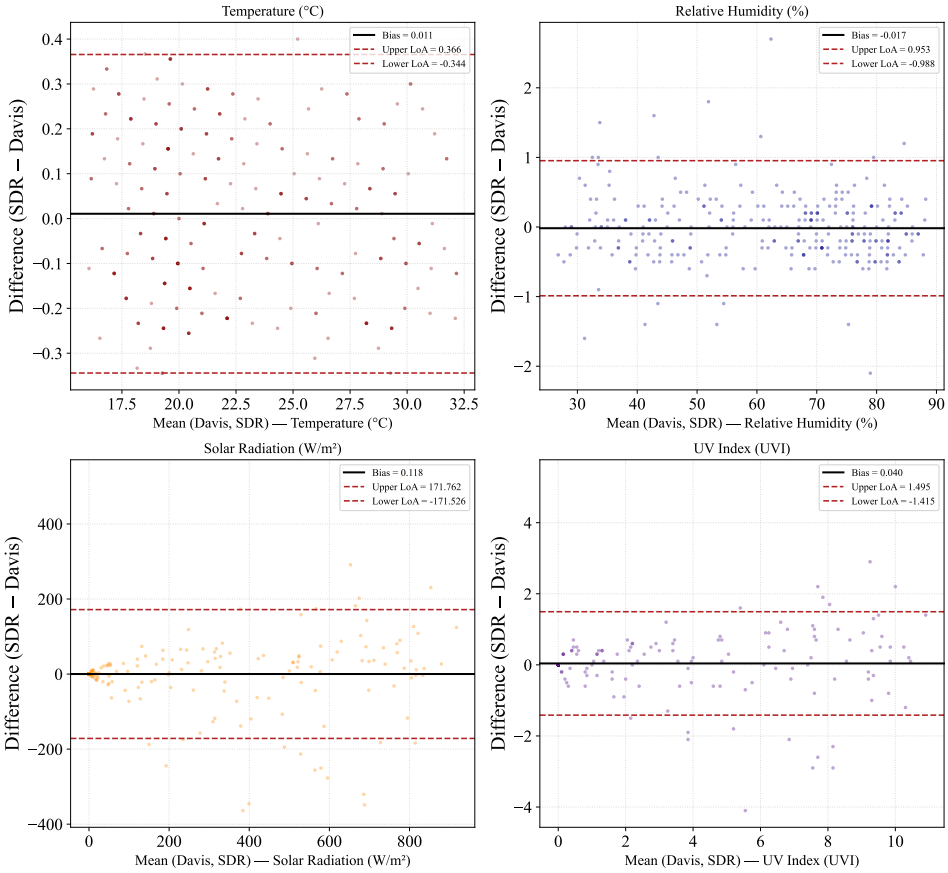


Figure 2 Bland–Altman agreement analysis between SDR-derived measurements and the Davis reference system for (a) air temperature, (b) relative humidity, (c) solar irradiance, and (d) UV index. The solid line indicates the mean bias and dashed lines represent the 95% limits of agreement. Temperature and relative humidity exhibit narrow agreement intervals and negligible bias, while broader limits are observed for solar irradiance and UV due to greater environmental variability and temporal aggregation effects.

Table 4 Summary of reliability and performance metrics obtained during the field evaluation.

Metric	Observed value
Evaluation period	29 Aug–4 Sep 2026
Validated database records	8,612
CRC-validated frames	$>$230,000
CRC-rejected frames	11,899
Packet delivery ratio	95.10%
Watchdog interventions	95
Mean time to recovery	1.0 s
Mean database-writing interval	60.20 s
Writing jitter	0.398 s
Empirical MTBF	90.95 min
Weibull shape parameter β	0.354
Weibull scale parameter η	8.06 min

4.5 Thermo-hygrometric behaviour

The decoded temperature and relative humidity followed a coherent thermo-hygrometric cycle throughout the six-day campaign. As shown in Figure 3, daily maxima exceeded 30 °C and night-time values fell to 16–18 °C. Figure 4 shows the complementary humidity response: values rose at night and fell during the warmest hours, frequently below 40%. The joint representation in Figure 5 makes the inverse relationship between the two variables explicit. This pattern matches expected atmospheric dynamics and provides qualitative evidence that the decoding preserved the physical structure of the signal over extended operation.

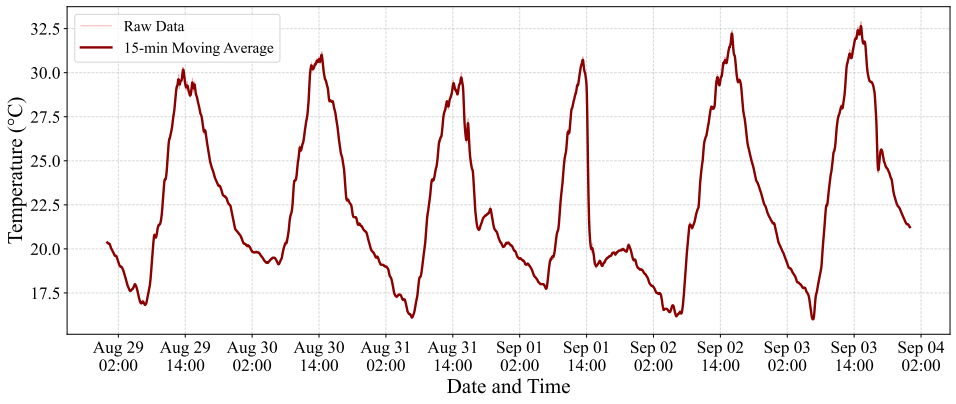


Figure 3 Multi-day air-temperature profile obtained from the SDR-based telemetry acquisition system during the six-day field campaign. Raw decoded observations are shown together with a 15-min moving average that highlights the diurnal thermal cycle. Daily maxima exceeded 30 °C and night-time minima remained near 16–18 °C.

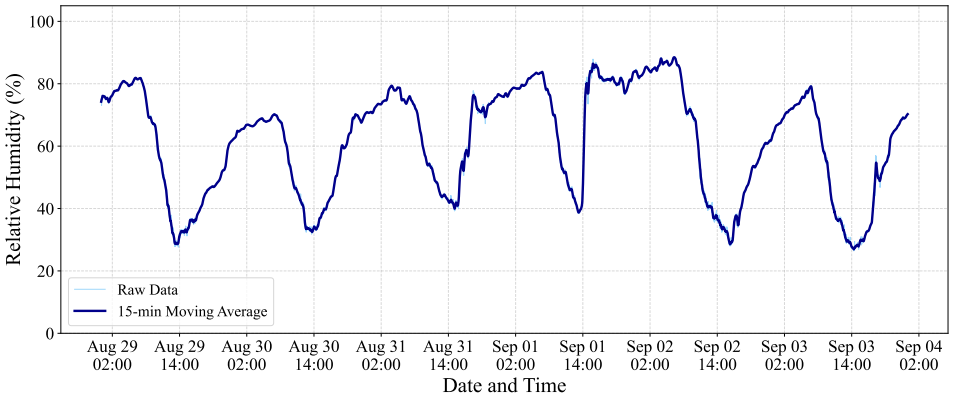


Figure 4 Multi-day relative-humidity profile obtained from the SDR-based telemetry acquisition system. The series shows the expected inverse daily relationship with air temperature, with elevated night-time humidity and pronounced daytime minima. The recurrence of this pattern over several days supports the physical coherence of the decoding workflow.

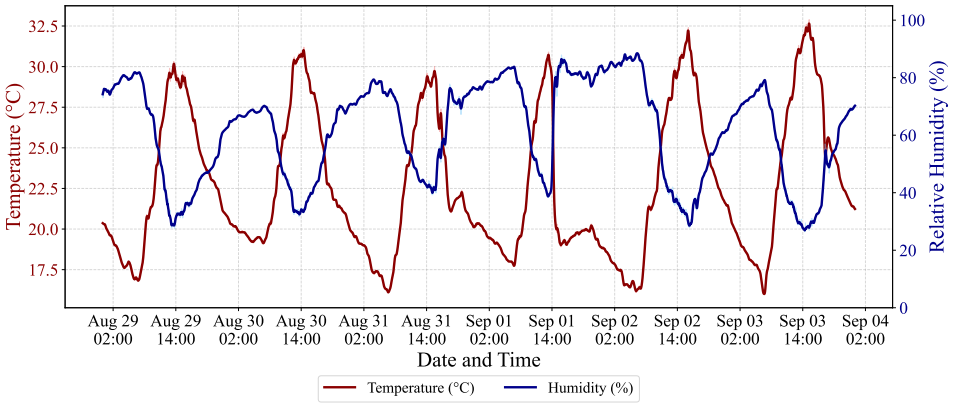


Figure 5 Simultaneous multi-day air-temperature and relative-humidity observations decoded from the SDR telemetry stream. The inverse relationship between the two variables agrees with expected atmospheric behaviour and indicates that payload interpretation, scaling factors and decoding routines operated coherently throughout the evaluation period.

4.6 Solar irradiance and UV index

The decoded solar irradiance and UV index followed consistent diurnal cycles over the six days of evaluation. As shown in Figure 6, daytime irradiance regularly exceeded 900 $\mathrm{W\,m^{-2}}$ and approached 1200 $\mathrm{W\,m^{-2}}$ under clear skies. Short depressions during daylight hours correspond to transient cloud passages. Figure 7 shows an almost identical temporal structure for the UV index, which rose after sunrise, reached midday maxima between about 10 and 13 UVI and returned to near-zero values at night. The combined representation in Figure 8 shows the close correspondence between the two variables, as expected for two quantities driven by the same incoming solar radiation. This agreement supports the bit-level extraction, scaling and decoding procedures applied to the optical telemetry fields.

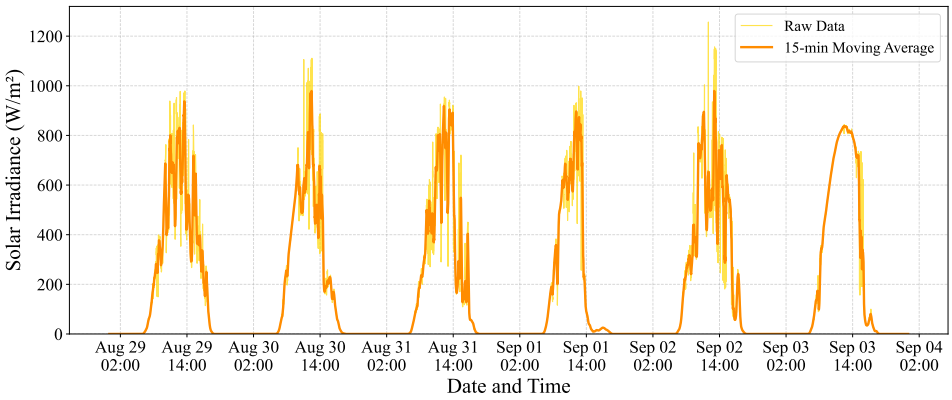


Figure 6 Multi-day solar-irradiance profile decoded from the SDR telemetry stream during the field campaign. Raw observations are shown together with a 15-min moving average. The daytime peaks and near-zero night-time values support the consistency of the optical-variable decoding workflow and reveal the influence of transient cloud cover on short-term irradiance.

4.7 System recovery and writing stability

The watchdog was activated 95 times during the campaign, with a mean recovery time of 1.0 s, that is, about 1.7% of the nominal one-minute writing interval. The accumulated downtime attributable to receiver stalls was therefore about 95 s over six days of operation, equivalent to an availability of 99.98%. Receiver stalls therefore seldom propagated to the persistence layer: the mean writing interval was 60.20 s with a jitter of 0.398 s, so periodic ingestion remained stable despite the recurrent restarts.

Figure 9 shows the temporal behaviour of the acquisition system. Watchdog activations recurred throughout the test, which indicates that receiver stalls were an operating characteristic of the low-cost SDR hardware rather than isolated incidents. The distribution of time-between-failure values combines short and long operating periods and yields an empirical MTBF of 90.95 min. Writing intervals remained grouped around the nominal 60 s period, so watchdog-triggered restarts were absorbed by the edge architecture without disrupting periodic persistence.

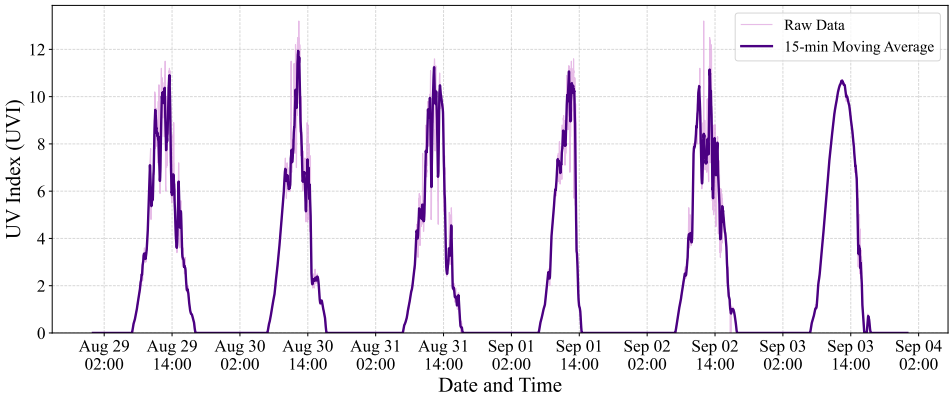


Figure 7 Multi-day UV-index profile reconstructed from the SDR-decoded telemetry packets. The daytime maxima and near-zero night-time baseline agree with expected atmospheric behaviour and support the bit-level extraction procedure applied to the UV telemetry field.

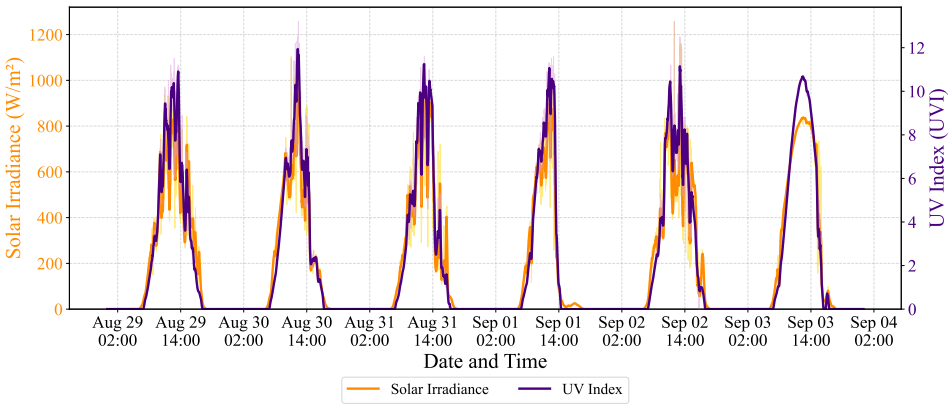


Figure 8 Simultaneous multi-day evolution of solar irradiance and UV index obtained from independently decoded telemetry fields. The two variables follow synchronised daily cycles, which indicates that the optical payload interpretation and scaling procedures remained internally consistent throughout the campaign.

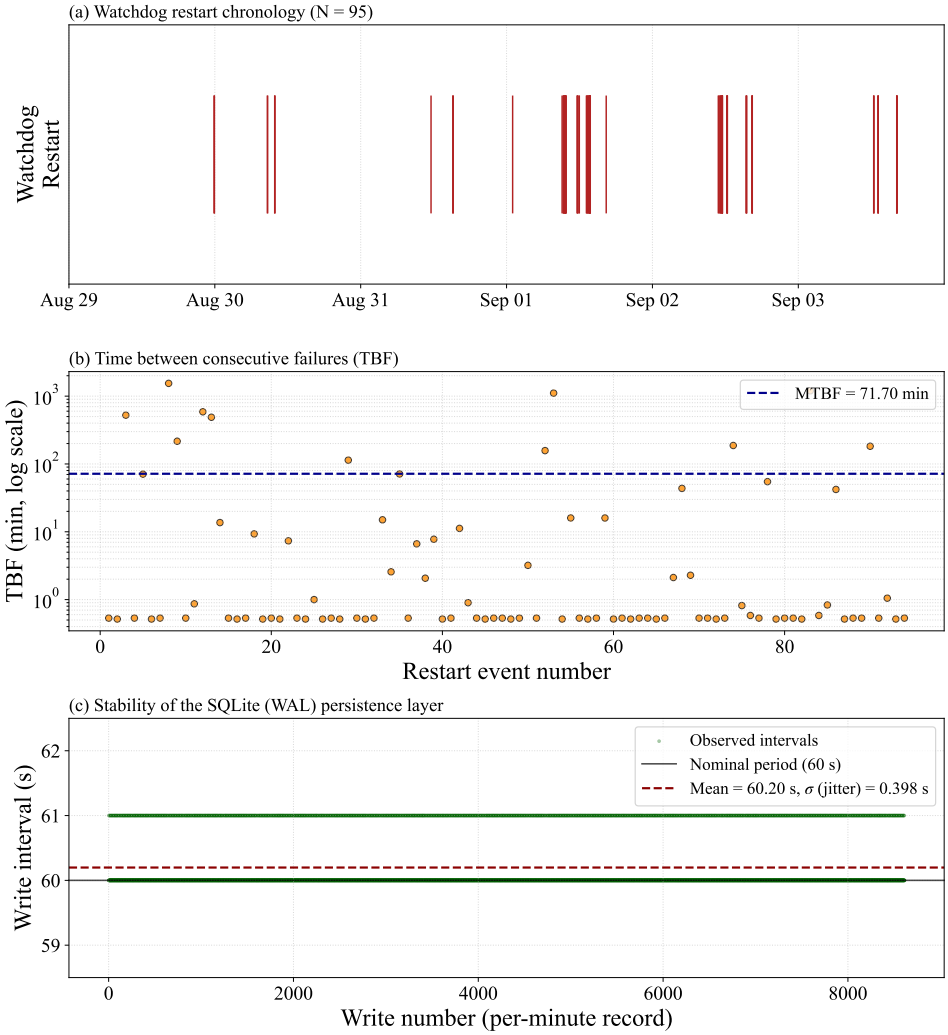


Figure 9 Reliability timeline and recovery behaviour of the SDR-edge acquisition system during field operation. (a) Temporal distribution of watchdog recovery events. (b) Time-between-failure (TBF) values for consecutive receiver stalls, together with the empirical mean time between failures (MTBF). (c) Stability of the persistence layer, showing the observed database-writing intervals relative to the nominal one-minute acquisition period. Receiver stalls occurred intermittently, but the watchdog restored operation within about one second and the writing interval remained close to its nominal value.

4.8 Weibull reliability interpretation

The maximum-likelihood Weibull fit of the times between receiver stalls produced a shape parameter $\beta = 0.354$ and a scale parameter $\eta = 8.06$ min. A value of β below unity denotes a decreasing failure rate, a pattern usually associated with early-life or initialisation-related failures rather than progressive hardware degradation. From a deployment perspective, low-cost SDR receivers are therefore expected to suffer recurrent interruptions during long unattended operation. Because the mean recovery time stayed close to one second, well below the acquisition interval, the watchdog absorbed most of the operational impact of these interruptions, which suggests that watchdog supervision is a design requirement rather than an optional safeguard for field-deployed SDR monitoring systems.

The fitted distribution and the empirical average describe different features of the same record and should be read together. With $\beta < 1$ the distribution is strongly right-skewed: its median is 2.9 min and its mean is 39.0 min, whereas the arithmetic mean of the observed intervals is 90.95 min. The difference arises because maximum-likelihood estimation is dominated by the numerous sub-minute intervals visible in Figure 9(b), while the arithmetic mean is raised by a small number of uninterrupted periods exceeding ten hours. Receiver stalls therefore occur in clusters separated by long stable stretches, rather than at a constant rate, which is the operational pattern the watchdog has to absorb.

5 Discussion

The contribution of this work is not the decoding of proprietary weather telemetry with low-cost SDR hardware, but the demonstration that such telemetry can be turned into a reliable, interoperable and fault-tolerant edge data layer for agricultural CPS. Previous agricultural monitoring studies have concentrated on sensing infrastructures, environmental measurements, communication architectures or cloud-based data management; the present work covers the whole acquisition pathway, from packet validation and payload decoding to edge persistence, automatic recovery and quantitative reliability assessment. This scope follows the evolution of agricultural CPS, in which sensing, communication, computation and decision support converge into a single digital infrastructure (An et al., 2017; Wang et al., 2006; Majumdar et al., 2021).

The validation results show that the acquisition workflow reproduced the manufacturer's interpretation of the sensor readings, and the Bland–Altman analysis places the differences between the two systems around a near-zero bias for every variable evaluated. Temperature and relative humidity gave near-unity correlations and biases of the order of 0.01 units, while solar irradiance and UV index also agreed closely despite the higher temporal variability of optical measurements. The packet interpretation logic, CRC validation, payload parsing, byte-order handling and scaling procedures therefore operated consistently. The purpose of the architecture is not to replace the meteorological sensors, but to provide an alternative telemetry-access path that allows integration with analytical, monitoring and decision-support platforms.

The main outcome is thus the conversion of proprietary meteorological telemetry into a validated, persistent and queryable edge data infrastructure. Where many SDR studies address radio experimentation, spectrum monitoring or communication research, this work traces a complete operational path from RF acquisition to agricultural decision support, which extends the role of SDR from research instrument to environmental-

information source within an edge-computing architecture. SDR has been applied to spectrum monitoring, environmental sensing and climate-related observation, but operational agricultural telemetry decoding and its integration into CPS infrastructures remain less explored (Machado and Wyglinski, 2015; Sorecau et al., 2023; Guevara et al., 2025; Salam et al., 2019).

The system can therefore be read as a bridge between proprietary sensing equipment and open agricultural information systems. Its relevance extends beyond weather monitoring, since the same architectural approach applies to other environmental-sensing platforms whose telemetry is inaccessible through open interfaces. Such interoperability matters for agricultural CPS deployments in which heterogeneous sensing infrastructures must coexist within shared decision-support and automation environments (An et al., 2017; Majumdar et al., 2021; Tzounis et al., 2017).

From the RF-acquisition perspective, a packet delivery ratio of 95.10% means that most intercepted frames were usable after CRC validation, and the rejected frames illustrate why integrity verification must precede ingestion into decision-support or control systems. Because greenhouse-drying decisions operate on timescales far slower than the telemetry transmission interval, occasional frame loss matters less than the guarantee that stored environmental states are valid, recent and internally consistent. The architecture consequently favours information reliability over exhaustive frame retention.

A further distinguishing feature of this study is the treatment of reliability as a design objective in its own right. Agricultural monitoring studies commonly report sensor accuracy, communication performance, connectivity or deployment success, whereas MTBF, MTTR, watchdog-recovery behaviour and Weibull failure modelling are seldom quantified. The present work therefore adds not only operational indicators but also a template for assessing long-term robustness in SDR-based agricultural telemetry. Existing agricultural reliability research has concentrated on fault detection, anomaly diagnosis and resilient sensing architectures rather than on the reliability of the telemetry-acquisition infrastructure itself (Shekarian et al., 2024; Alotaibi, 2026).

The reliability analysis indicates that low-cost SDR hardware cannot be assumed stable during extended unattended operation. An empirical MTBF of 90.95 min and a Weibull shape parameter below unity point to recurrent interruptions of the early-life or initialisation type rather than progressive degradation. The recovery architecture reduced the operational consequence of these interruptions: an MTTR of about 1.0 s is far shorter than the one-minute persistence interval, so most stalls were absorbed before they produced data gaps. Operational continuity thus emerged from the software architecture around the receiver, not from the receiver itself, which argues for treating self-healing mechanisms as an architectural requirement in agricultural deployments of low-cost SDR hardware. This reading is consistent with recent discussions of resilience, fault tolerance and dependable operation in long-term agricultural IoT (Gong et al., 2025; Alotaibi, 2026; Shekarian et al., 2024).

From an agricultural-management perspective, the decoded variables supply the external context needed for proactive drying decisions. Relative humidity determines whether ventilation will remove or add moisture in the chamber, rain information can inhibit vent opening, solar irradiance indicates the natural drying potential available, and wind speed and direction inform ventilation-risk assessment. The system should therefore be understood as a decision-support acquisition layer that feeds external environmental information to future closed-loop drying controllers (Sacaleanu et al., 2024; Yousaf et al., 2022; Petraki et al., 2025).

The multi-day profiles add qualitative evidence on the physical consistency of the decoding. Temperature and relative humidity showed the expected inverse diurnal relationship, and solar irradiance and UV index showed matching daytime cycles with near-zero night-time baselines. These patterns complement the quantitative agreement metrics and indicate that the telemetry interpretation remained coherent over prolonged operation.

Several limitations should be acknowledged. The evaluation used a single weather station, one SDR receiver and one deployment site. The operating period covered six days, which supports a proof of concept but not a seasonal reliability assessment. The validation addressed agreement with the manufacturer's interpretation rather than metrological accuracy against independently calibrated instruments. Finally, drying actuation was not included, so the effect of the acquired telemetry on drying performance remains to be quantified.

Within these limits, the architecture is reproducible and scalable: it relies on low-cost SDR hardware, an ARM/Linux edge platform, open software components, explicit integrity validation and a standard embedded SQL archive. It should not be read as a replacement for commercial weather stations, but as an interoperability and resilience layer that brings proprietary environmental telemetry into open agricultural CPS infrastructures. The results indicate that low-cost SDR platforms support robust field operation when they are combined with validation, persistence and self-recovery mechanisms, which offers a practical path towards affordable and reproducible agricultural cyber-physical systems (An et al., 2017; Machado and Wyglinski, 2015; Gong et al., 2025).

6 Conclusions

Summary of Findings

This study presented and evaluated a low-cost SDR-based edge architecture for the real-time acquisition, validation, decoding and persistence of proprietary weather telemetry in agricultural cyber-physical systems. Comparison with the Davis reference system showed that the decoding workflow reproduced the environmental information carried by the proprietary telemetry channel, with mean absolute errors of 0.156 °C for air temperature and 0.352% for relative humidity. Over six days of field operation the system generated 8,612 validated records, with a packet delivery ratio of 95.10%, a mean time to recovery of 1.0 s, a mean writing interval of 60.20 s with 0.398 s jitter, and an empirical MTBF of 90.95 min after 95 watchdog-triggered restarts.

Scientific Contribution

The architecture brings interoperability, edge computing, data persistence, watchdog-based recovery and quantitative reliability assessment into a single operational framework. By combining CRC-based packet validation, asynchronous processing, redundant storage and self-recovery, it converts proprietary environmental telemetry into a structured and continuously available data layer for agricultural CPS. The reliability analysis further showed that operational continuity depended on architectural resilience rather than on the intrinsic stability of the receiver, which places self-healing mechanisms among the design requirements of long-term SDR deployments. To the best of the authors' knowledge, few agricultural telemetry studies have combined SDR reception, proprietary

telemetry interoperability, edge persistence, watchdog-driven fault recovery and quantitative reliability modelling within one monitoring architecture.

Future Work

Future work will integrate the external weather-information layer with internal greenhouse sensors, actuator feedback and automated control strategies for drying. Longer deployments covering seasonal conditions are needed to assess robustness and scalability, and validation against independently calibrated instruments will establish metrological accuracy beyond decoding fidelity. Publishing the datasets, software components and reliability-monitoring workflows should also support reproducible research on affordable and dependable agricultural cyber-physical infrastructures.

Author Contributions

Declarations

Data and Code Availability

The source code, processed datasets, figure-generation scripts and unit tests are publicly available at https://github.com/Sibajx/davis-sdr-greenhouse (Ramírez-Sibaja et al., 2026). The repository contains the acquisition daemon, the CRC-16 validation and decoding routines, the cleaned one-minute dataset used for the figures reported here, the reliability-analysis scripts and a representative excerpt of the operation log; the complete log is omitted because of its size and redundancy. The full operation log and the raw hexadecimal captures are available from the corresponding author on reasonable request.

Ethical and Legal Interoperability Statement

The SDR decoding workflow presented in this study was developed for academic research, for interoperability and for the integration of locally owned environmental sensing equipment into a custom agricultural CPS. The system passively receives weather telemetry generated by the authors' own station and is not intended to interfere with, disrupt or compromise any third-party communication system. Researchers who replicate this workflow should verify the applicable radiofrequency regulations and intellectual-property constraints before field deployment.

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Conflicts of Interest

The authors declare no conflict of interest.

Use of AI Technology

The authors used Claude (Anthropic), a large language model, during the preparation of this manuscript. Its use was limited to language editing and editorial review of text written by the authors: improving clarity and concision, unifying spelling and terminology, reducing redundancy between sections, checking the internal consistency of the reported figures across text, tables and captions, and formatting bibliographic and software references.

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